Chaitanya Bharathi Institute of Technology (CBIT)

Department of Computer Science and Engineering

CASE STUDY 2: CLUSTERING ANALYSIS

Customer Segmentation on East-West Airlines Data Using K-Means, Hierarchical, and DBSCAN Clustering

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Section:	CSE-3
Academic Year:	III Year, V Semester (2026-27)
Course / Subject:	Fundamentals of Data Science (FDS)
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Date of Submission:	26-10-2026

1. Problem Statement and Objectives

East-West Airlines maintains an extensive frequent flyer loyalty programme. Understanding different customer segments based on flight frequency, mileage accumulation, and credit card partner usage is essential to optimize marketing spend and design targeted retention offers. The objective of this case study is to develop an end-to-end unsupervised learning pipeline to identify distinct customer groups from behavioural data.

Key objectives include: (1) cleaning and transforming highly skewed continuous features, (2) detecting and handling outliers without discarding genuine high-tier accounts, (3) evaluating three fundamentally distinct clustering paradigms (partitioning via K-Means, agglomerative hierarchical clustering with multiple linkages, and density-based clustering via DBSCAN), (4) comparing algorithm performance using internal validity indices (Silhouette, Davies-Bouldin, and Calinski-Harabasz), and (5) formulating actionable business strategies tailored to each identified customer persona.

2. Dataset Description

The dataset originates from Shmueli and Bruce (2016) and contains frequent flyer records for 3,999 customers across 12 features. The variables capture account age, balance, tier qualifying miles, mileage earned through partner credit cards (cc1, cc2, cc3), non-flight bonus transactions, flight transactions in the past 12 months, and award redemption history.

Pre-modelling inspection confirmed 0 missing values and 0 duplicate records. However, substantial data quality challenges were present: the mileage and transaction attributes exhibit severe positive skewness (e.g. cc3_miles skewness = 17.20, Balance skewness = 5.00). Standard IQR outlier filtering on raw features would have discarded 569 records (14.2%) in Flight_miles_12mo and 266 (6.7%) in Balance, erroneously eliminating the airline's most valuable elite flyers. Customer ID was dropped as a non-informative identifier, and the Award? binary indicator was excluded from cluster distance matrices and retained strictly for post-clustering profiling.

3. Methodology

To establish a robust clustering framework, features with skewness > 1.0 were transformed using np.log1p (log(x + 1)), which compresses extreme ranges while preserving zeros. Outlier removal was subsequently applied using a standard Z-score threshold of |z| > 3 on the log-transformed space, removing 376 extreme multivariate outliers (9.40%) and retaining 3,623 records. All 10 feature columns were then normalized using StandardScaler (zero mean, unit variance).

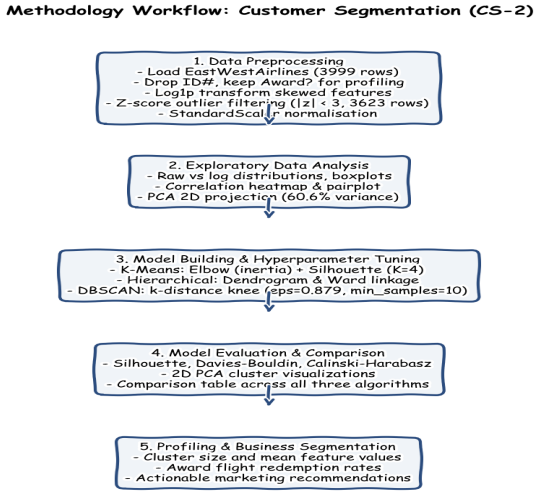


Figure 1: End-to-end customer segmentation methodology workflow

Model building evaluated three paradigms: K-Means (tested for K=1 to 10 with 10 random restarts), Agglomerative Hierarchical clustering (evaluating Ward, Complete, Average, and Single linkages), and DBSCAN (parameterized via 5-nearest neighbour k-distance knee analysis and a grid search across eps and min_samples).

4. Statistical Findings & Exploratory Data Analysis

Skewness analysis identified 8 features with severe right-skewness > 1.0: cc3_miles (17.20), cc2_miles (8.73), Qual_miles (7.52), Flight_miles_12mo (7.45), Flight_trans_12 (5.49), Balance (5.00), Bonus_miles (2.84), and Bonus_trans (1.16). Applying np.log1p transformed these distributions toward near-normal symmetry, stabilizing Euclidean distance calculations.

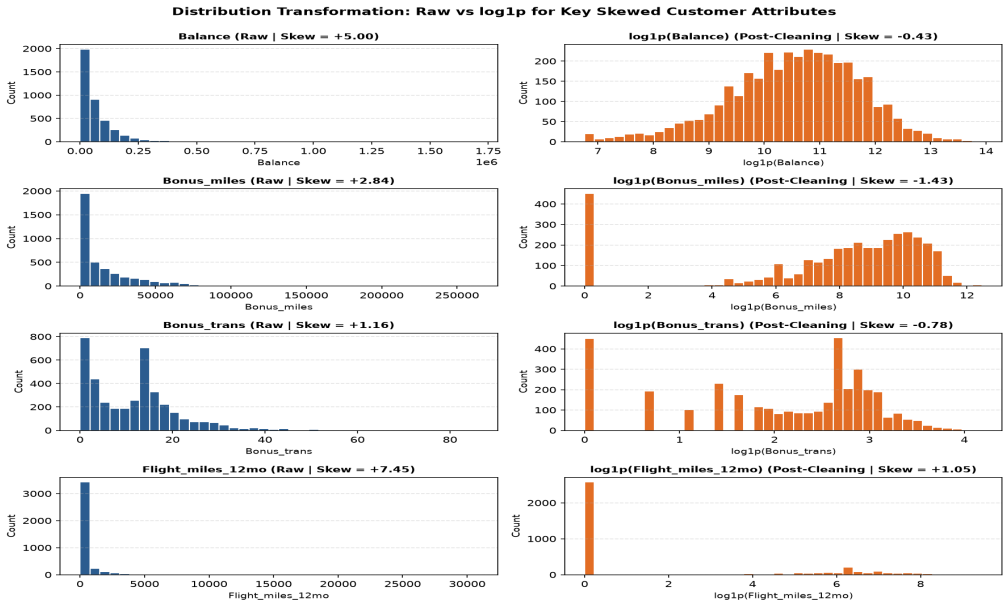


Figure 2: Distribution transformation: Raw vs log1p for key continuous attributes showing skewness reduction

Correlation analysis indicated strong collinearity between credit card miles and bonus mileage (r = 0.62), as well as between recent flight miles and flight transaction counts (r = 0.93), and bonus miles and bonus transactions (r = 0.90). Constant features (cc2_miles and cc3_miles) have zero variance post-filtering and are excluded from the correlation matrix to ensure a clean, fully populated structure. PCA revealed that the first two components capture 39.73% and 20.83% of total variance respectively (60.56% cumulative).

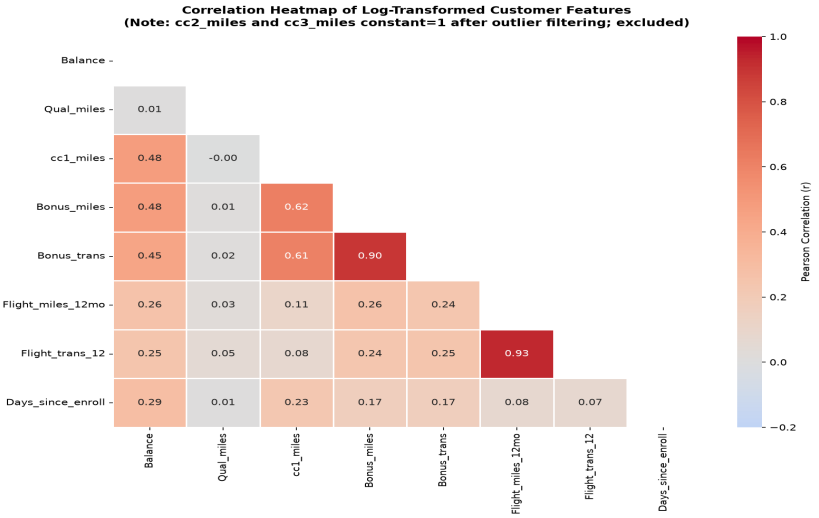


Figure 3: Feature correlation matrix across log-transformed customer attributes

5. Model Evaluation and Algorithm Comparison

K-Means hyperparameter selection combined inertia elbow analysis and silhouette score evaluation across K=2 to 10. Inertia declined sharply from K=1 (28,984) to K=4 (12,568), demonstrating an elbow point at K=4. Silhouette scores reached an absolute maximum at K=4 (0.3199), confirming 4 clusters as the optimal partitioning choice.

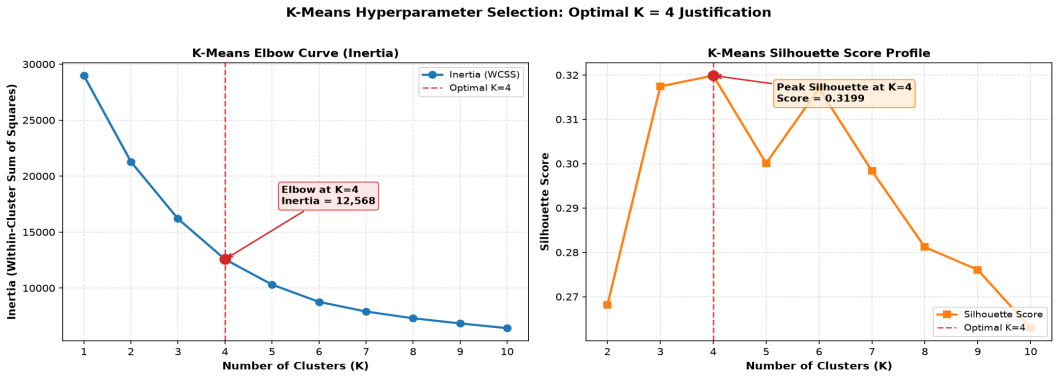


Figure 4: K-Means evaluation: Inertia elbow curve (left) and Silhouette score profile (right) with K=4 highlighted

For Hierarchical clustering, truncated dendrograms were constructed across four linkage criteria. Ward linkage achieved the highest silhouette score (0.3317) at K=4 with clean, balanced branches. Single linkage exhibited severe chaining (silhouette = 0.2641), while Complete (0.2453) and Average (0.2518) produced less coherent partitions.

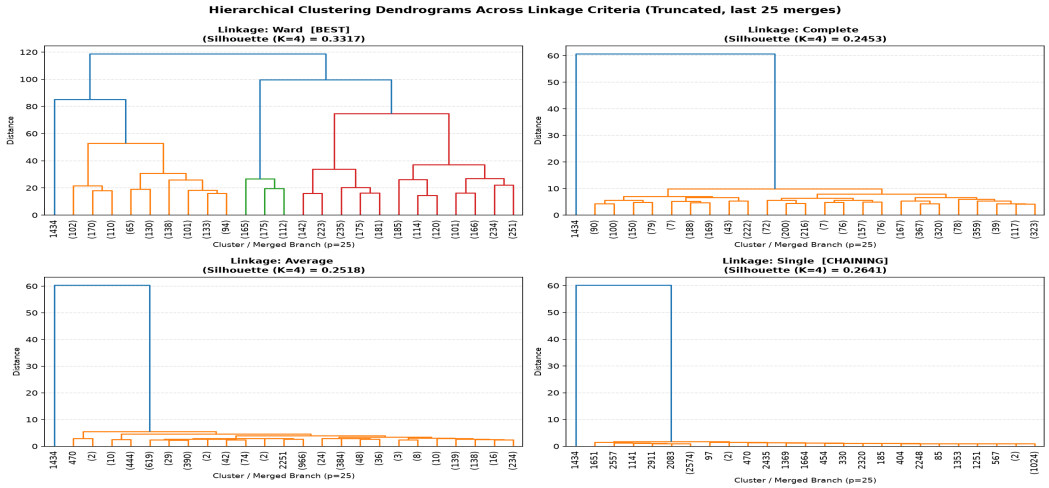


Figure 5: Truncated hierarchical clustering dendrograms across linkage criteria showing silhouette at K=4

For DBSCAN, the 5-nearest neighbour distance curve sorted in ascending order demonstrated a clear knee region between distances 0.88 and 0.94. Grid search across eps and min_samples identified eps = 0.878 and min_samples = 10 as optimal, separating 220 accounts (6.07%) as noise and identifying 3 dense customer cores with non-noise silhouette = 0.3353.

Table 1: Clustering Algorithm Performance Comparison

Algorithm	Clusters	Noise Pts (%)	Silhouette	Davies-Bouldin	Calinski-Harabasz
K-Means	4	0 (0.0%)	0.3199	0.9021	1575.63
Hierarchical (ward)	4	0 (0.0%)	0.3317	0.8735	1410.12
DBSCAN	3	220 (6.07%)	0.3353	1.1554	1512.32

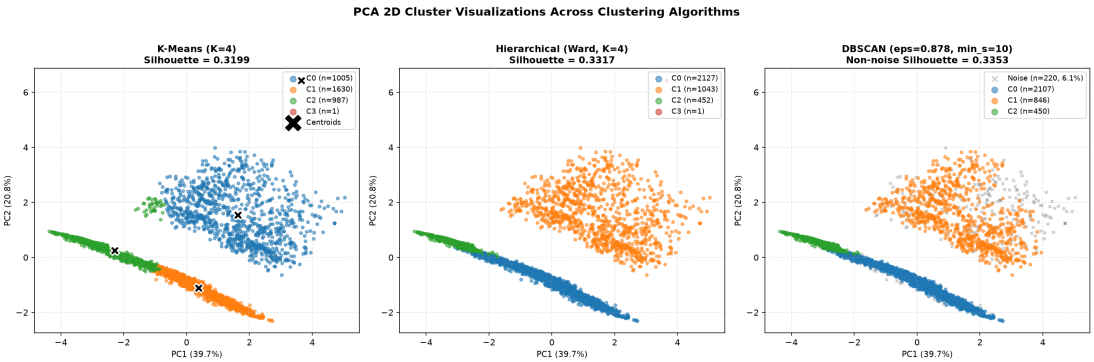


Figure 6: 2D PCA cluster visualizations with centroids (K-Means) and noise classification (DBSCAN)

6. Cluster Profiling and Marketing Strategy

K-Means segments were profiled by unscaling cluster centroids back to original units and analyzing award redemption rates:

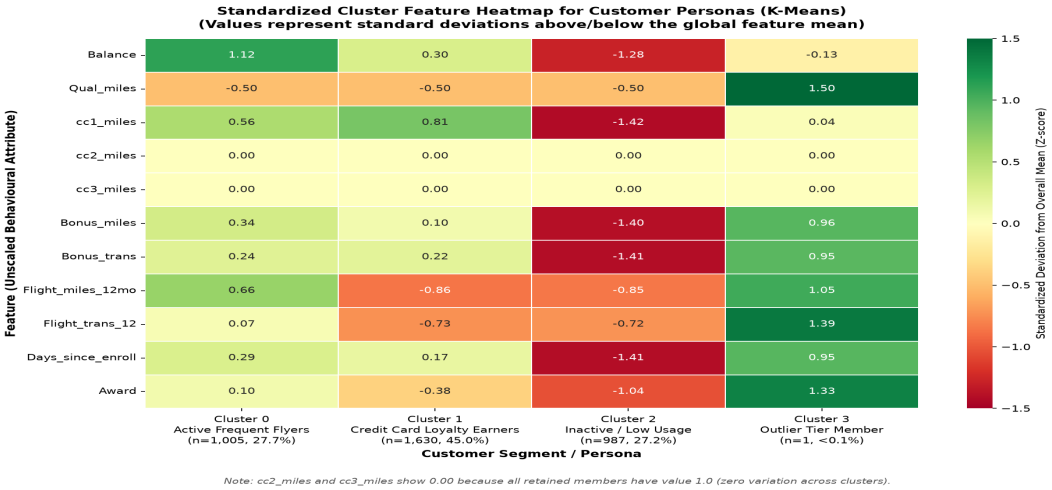


Figure 7: Standardized cluster feature heatmap for K-Means customer personas

Cluster 0: Active Frequent Flyers (1,005 members, 27.7%)
Profile: High average balance (103,836 miles), 1,076 flight miles in the past 12 months, 3.03 flight transactions, and 54.2% award flight redemption rate. Strategy: Retain with tier upgrades, premium lounge access, and partner travel perks.

Cluster 1: Credit Card Loyalty Earners (1,630 members, 45.0%)
Profile: Substantial balance (75,965 miles), 20,900 bonus miles through co-branded credit cards (cc1 tier = 2.5), but zero flights recorded in the past year; 36.2% award rate. Strategy: Convert mileage into revenue flights via limited-time seat discounts and redemption bonus windows.

Cluster 2: Inactive / Infrequent Members (987 members, 27.2%)
Profile: Low account balances (22,418 miles), minimal transactions (1.79 bonus transactions), 10 flight miles, and only 11.8% award rate. Strategy: Low-cost automated email re-engagement campaigns with welcome-back bonus incentives to prevent churn.

Cluster 3: Outlier Tier Account (1 member, <0.1%)
Profile: Account with extraordinary qualification miles (250 Qual_miles, 8 flight transactions) isolated as a distinct boundary point.

7. Conclusion and Limitations

This investigation demonstrated that customer segmentation on airline frequent flyer data is highly effective when combined with appropriate transformations. K-Means and Hierarchical Ward clustering yielded consistent, highly interpretable customer partitions. DBSCAN demonstrated utility in isolating 220 noise accounts with atypical usage patterns.

Key limitations include: (1) K selection retains an element of subjectivity despite elbow and silhouette validation, (2) clusters depend on log1p scaling and StandardScaler assumptions, (3) DBSCAN struggles with varying density across features, and (4) PCA 2D projections capture 60.6% of variance, omitting higher-dimensional variation.

8. References & Attribution

1. Shmueli, G., and Bruce, P. C. (2016). *Data Mining for Business Analytics: Concepts, Techniques, and Applications in Python*. John Wiley & Sons. East-West Airlines dataset.
2. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
3. GitHub Repository: https://github.com/katthi-goutham-reddy/FDS-case-study

Attribution: Code structure drafted with AI assistance (Claude); executed, verified and interpreted by Katthi Goutham Reddy. Dataset source: Shmueli and Bruce (2016), East-West Airlines frequent flyer dataset.